

The explanation of each stage is in the handwritten page .

STAGE 1:

```
javierub@DESKTOP-CHBPSS9$ ./echo_server
server: listening on port 12345
server: client connected from 127.0.0.1:50598
server: received 1024 bytes
server: received 1 bytes
server: client disconnected
```

```
javierub@DESKTOP-CHBPSS9$ ./echo_client 127.0.0.1 12345 1025
client: 1025 bytes echoed in 0.254 ms (identical)
```

STAGE 2:

Size	Average time (ms)	Server Chunks
10	0.382	1
1,000	0,376	1
10,000	2,045	10
100,000	25,469	98

STAGE 3:

I added "sleep(5)" when the server reads the client data to simulate the tests easier.

TEST 1:

```
javierub@DESKTOP-CHBPSS9$ ./echo_server
server: listening on port 12345
server: client connected from 127.0.0.1:42602
server: received 1024 bytes
^C
```

```
javierub@DESKTOP-CHBPSS9$ ./echo_client 127.0.0.1 12345 1025
read: Connection reset by peer
```

TEST 2:

```
javierub@DESKTOP-CHBPSS9$ ./echo_client 127.0.0.1 12346 1025
connect: Connection refused
```

TEST 3:

```
javierub@DESKTOP-CHBPSS9$ ./echo_server
```

```
server: listening on port 12345
```

```
server: client connected from 127.0.0.1:40104
```

```
server: received 1024 bytes
```

```
server: received 1 bytes
```

```
server: client disconnected
```

```
server: client connected from 127.0.0.1:40098
```

```
server: received 1024 bytes
```

```
server: received 1 bytes
```

```
server: client disconnected
```

```
javierub@DESKTOP-CHBPSS9$ ./echo_client 127.0.0.1 12345 1025 &
```

```
./echo_client 127.0.0.1 12345 1025 &
```

```
wait
```

```
[1] 4391
```

```
[2] 4392
```

```
client: 1025 bytes echoed in 10001.340 ms (identical)
```

```
[1]- Done          ./echo_client 127.0.0.1 12345 1025
```

```
client: 1025 bytes echoed in 20002.835 ms (identical)
```

```
[2]+ Done          ./echo_client 127.0.0.1 12345 1025
```